Customer Behaviour & Revenue Optimization Using RFM Segmentation

Data Intelligence | Analytics | Business Strategy

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Executive Summary

This case study analyzes customer behavior and sales performance using transactional retail data to identify revenue concentration, detect churn risk, and uncover opportunities for customer retention and revenue optimization.

Using RFM (Recency, Frequency, Monetary) segmentation, customers are grouped based on purchasing behavior. The analysis reveals that a small segment of high-value customers contributes a disproportionate share of revenue, while other segments present both growth opportunities and retention risks. The findings support targeted marketing strategies, pricing optimization, and operational improvements.

Problem Statement

Businesses often struggle to identify high-value customers and detect churn risk. Without effective segmentation, marketing efforts become inefficient, leading to lost revenue and reduced profitability.

Objective

- Identify high-value customers driving revenue
- Detect customers at risk of churn
- Understand customer purchasing behavior
- Provide actionable recommendations for revenue optimization

Data Overview

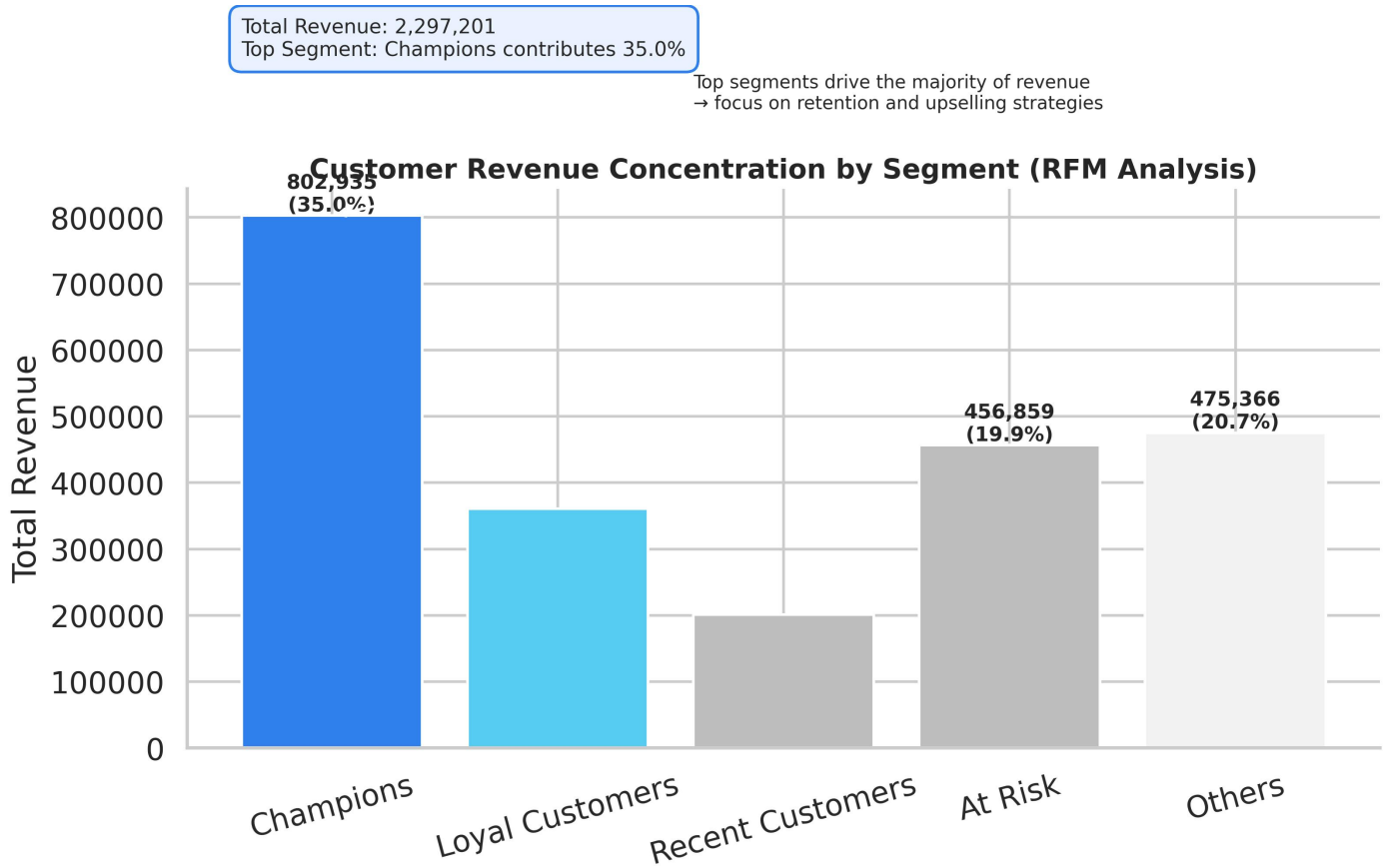
The Superstore dataset includes transactional data such as:
- Customer orders
- Sales and profit
- Discounts applied
- Shipping details

This dataset provides a comprehensive view of customer behavior across segments and regions.

Methodology

- Data Cleaning and preparation of transactional records
- Exploratory Data Analysis (EDA) to identify trends and patterns
- Feature Engineering (shipping delay, time-based metrics)
- RFM Segmentation to classify customers based on behavior
- Insight Generation to identify revenue drivers and risks
- Visualization for executive-level communication

Customer Revenue Analysis



Key Insights

- Revenue is highly concentrated among high-value customers, with Champions contributing approximately 35% of total revenue
- At Risk and Other segments represent both revenue vulnerability and untapped growth opportunities
- Discounts negatively impact profitability in multiple cases
- Shipping delays may reduce customer satisfaction and retention
- Revenue follows the Pareto principle, with a small group driving the majority of value

Business Impact

- Enables targeted retention and upselling strategies
- Identifies churn risk to prevent revenue loss

• Supports optimization of pricing and discount strategies
• Improves operational efficiency and customer experience

Recommendations

• Prioritize retention strategies for high-value customers (Champions)
• Implement targeted campaigns for At Risk customers
• Optimize discount strategies to protect margins
• Improve logistics performance to reduce shipping delays

Conclusion

This case study demonstrates how RFM segmentation transforms raw transactional data into actionable business intelligence.

By identifying key customer segments and their contribution to revenue, organizations can make data-driven decisions, improve customer retention, and drive sustainable growth.